

PREDICTION OF GOLD PRICE IN INDIA USING MACHINE LEARNING



1. INTRODUCTION

- Throughout the history of mankind, Gold has been money. Gold was money. Gold is money.
- Gold is one of the most highly desired and useful metals in the world. Arguably no other metal has been held in higher esteem throughout history, with almost every established culture using gold to symbolize power, beauty, purity, and accomplishment.
- Today, gold still occupies an important place in our culture and society.
- As per statistics, about 80% of gold is converted into jewelry. It is the most standard use of gold and is common amongst all cultures.

1.1 Importance Of Gold

- Gold is a unique asset: highly liquid, yet scarce; it's a luxury good as much as an investment. Gold is no one's liability and carries no counterparty risk. As such, it can play a fundamental role in an investment port. Gold has been a highly preferred investment tool because of the liquidity it offers. In comparison to other forms of investment like stocks and bonds, gold has proven to be an asset that is easy to liquidate, and thus, in times of an emergency, it can easily act as an investment cushion.
- The metal is abundant enough to create coins but rare enough so that not everyone can produce them. Gold doesn't corrode, providing a sustainable store of value, and humans are physically and emotionally drawn to it. Societies and economies have placed value on gold, thus perpetuating its worth.
- Gold is a currency. Gold has a price, and that price will fluctuate relative to other forms of exchange, such as the U.S. dollar, the euro, and the Japanese yen. Gold can be bought and stored, but it is not usually used directly as a method of payment. However, it is highly liquid and can be converted to cash in almost any currency with relative ease.

1.2 Why Gold Price Prediction?

- Among all precious metals, gold might be the most popular choice for investment.
- Gold prices change on daily basis and are also affected by major world events.
- Gold is an important saving and investment instrument in India, it is expected that gold may be looked upon as an alternative asset for those holding idle money and for speculative purposes.
- In the rising inflation rate, money loses its value gradually, and therefore, people tend to buy physical assets like, gold which will retain their value.
- By analyzing financial, geopolitical, and global data, it is possible to predict future fluctuations in the price of gold, which will allow investors to improve the management of their portfolios in the face of unexpected movements of the market.
- Maintaining gold reserves by central banks is crucial to support the current economic structure of the world. Some major companies and investors also invest a huge amount of money in gold. Although not easy, predicting the rate of gold would help investors and central banks to better decide when to sell and buy it, thus maximizing their profits.

2.OBJECTIVE

- To predict the gold prices in India by analysing the daily historical data of the different factors affecting it.

3. DATA SNIPPET

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		Date	SILVER	USD	INR	LBMA	CRUDE	HUI	S&P	SENSEX	NIFTY	PRICE	
2	0	01-01-2012											
3	1	02-01-2012	24.26		ND		69.12			15517.92		47179	
4	2	03-01-2012	24.18	79.61	53.11	1590	69.3	522.2	1304.28	15939.36	4765.3	47612	
5	3	04-01-2012	24.06	80.09	52.85	1603		521.44	1301.78	15882.64	4749.65	47584	
6	4	05-01-2012	24.37	80.94	52.89	1614.5	68.33	522.64	1295.09	15857.08	4749.95	47158	
7	5	06-01-2012	24.74	81.24	52.67	1621	66.56	517.66	1296.06	15867.73	4754.1	47169	
8	6	07-01-2012								15848.8		47132	
9	7	08-01-2012	24.81				68.26						
10	8	09-01-2012	24.8	81.05	52.39	1618		519.76	1292.99	15814.72	4742.8	47280	
11	9	10-01-2012	23.92	80.8	51.59	1627	69.1	527.76	1298.82	16165.09	4849.55	47225	
12	10	11-01-2012	24.22	81.35	51.78	1641	68.19	522.03	1297.52	16175.86	4860.95	46940	
13	11	12-01-2012	24.01	80.77	51.51	1652.5		526.74	1302.3	16037.51	4831.25	46363	
14	12	13-01-2012	24.01	81.43	51.43	1642		519.39	1297.48	16154.62	4866	46388	
15	13	14-01-2012										45962	
16	14	15-01-2012	24.15										
17	15	16-01-2012	24.11		ND	1643.5				16189.36	4873.9	45886	
18	16	17-01-2012	23.6	81.18	50.69	1662		511.51	1295.43	16466.05	4967.3	46640	
19	17	18-01-2012	23.82	80.6	50.4	1657		512.93	1285.58	16451.47	4955.8	48450	
20	18	19-01-2012	23.94	80.21	50.3	1664		502.07	1268.21	16643.74	5018.4	48111	
21	19	20-01-2012	23.66	80.22	50.22	1646		501.57	1266.74	16739.01	5048.6	47817	
22	20	21-01-2012										48045	
23	21	22-01-2012	22.95										
24	22	23-01-2012	23.11	79.69	49.99	1675		508.89	1271.81	16751.73	5046.25	48032	
25	23	24-01-2012	23.23	79.86	50.02	1669		497.85	1265.95	16995.77	5127.35	48281	
26	24	25-01-2012	23.42	79.57	50.18	1659		530.59	1271.48	17077.18	5158.3	47577	
27	25	26-01-2012	23.66	79.41	49.42	1713		535.78	1275.77				
		Final Data											

4. DATA DESCRIPTION

Data used for this study is collected from 1st January 2012 to 31st August 2021 from various sources.

- The variables considered in the data include:
 - HUI Gold Index
 - Standard and Poor's (S&P) 500 index
 - LBMA Silver Price
 - LBMA Gold Price
 - Crude Oil Price
 - Sensex
 - Nifty
 - INR
 - USD

Variables	Currency/Country	Unit
LBMA Gold Price	US Dollar	Per Troy Ounce
LBMA Silver Price	US Dollar	Per Ounce
Crude Oil Price	US Dollar	Per Barrel
HUI Index	US	
S&P Index	US	
BSE Sensex	India	
NSE Nifty	India	
USD Index	US	
INR	Indian Rupee	Per US Dollar
Gold Price	Indian Rupee	Per 10 Grams

df.info ()

```

#      Column      Non-Null Count  Dtype
---  -
0      Unnamed: 0    3531 non-null     int64
1      Date          3531 non-null     object
2      Silver        3008 non-null     float64
3      USD           2413 non-null     float64
4      INR           2416 non-null     float64
5      LBMA          2443 non-null     float64
6      Crude         2440 non-null     float64
7      HUI           2432 non-null     float64
8      S&P           2431 non-null     float64
9      Sensex        2395 non-null     float64
10     Nifty          2363 non-null     float64
11     Price          2586 non-null     float64
dtypes: float64(10), int64(1), object(1)

```

df.describe()

	Unnamed: 0	Silver	USD	INR	LBMA	Crude	HUI	S&P	Sensex	Nifty	Price
count	3531.000000	3008.000000	2413.000000	2416.000000	2443.000000	2440.000000	2432.000000	2431.000000	2395.000000	2363.000000	2586.000000
mean	1765.000000	20.546894	91.273469	65.278386	1411.677569	65.306709	240.792796	2398.106734	30236.320129	9116.127275	32532.238592
std	1019.456228	5.184134	7.103653	6.524312	232.938434	22.724425	96.040156	738.363610	9185.897164	2701.116413	7186.362978
min	0.000000	11.978000	78.269997	48.650000	1050.600000	-36.980000	100.700000	1277.060000	15488.370000	4742.800000	24597.000000
25%	882.500000	16.763250	82.800003	61.547500	1239.325000	48.127500	178.077500	1884.020000	23532.075000	7219.000000	28136.000000
50%	1765.000000	18.549000	93.669998	65.705000	1313.600000	58.790000	212.825000	2181.300000	28361.100000	8685.900000	29868.000000
75%	2647.500000	23.602500	97.010002	70.782500	1608.625000	88.735000	273.427500	2843.115000	36357.995000	10890.950000	32017.500000
max	3530.000000	37.140000	103.290001	76.950000	2061.500000	110.620000	551.240000	4528.790000	57257.980000	16931.050000	55845.000000

5. DATA PREPARATION

There were no missing values for the data except the government holidays & the weekends.

All the null values have been removed from the data. Before removing the null values, we had 3531 rows x 12 columns & after removing the null values we have a total of 2220 rows x 12 columns.

Since the data is daily, the values of the variables taken in the study fluctuate every second, so due to that there were many outliers in the data and removing them could lead to loss of data. So, the outliers were not removed from the data.

6. FACTORS AFFECTING GOLD PRICE

6.1 Inflation

- When there is a rise in inflation in the country, the value of rupee goes down and people save money in the form of gold.
- Therefore, if a high level of inflation sustains over a long period of time, then gold becomes an instrument to hedge against inflationary factors when gold prices go up.

6.2 Global Market Force

- If global gold price increase, India's import price increase which reflects in domestic market cost of gold.
- The prices of gold for the world are set by the **London Bullion Market Association (LBMA)**.

6.3 Government Gold Reserves

- When central banks of large countries start holding gold reserves and procuring more gold, the price of gold goes up.
- This is because the flow of cash in the market is increased while the supply of gold goes down.

6.4 Supply & Demand

It costs more to get less gold.

Thus, the increase in cost of gold production sometimes results in higher gold prices.

6.5 Interest Rates

- When the interest rates of banks are lowered, more people borrow more loans which leads to more money in the economy.
- Thus, consumption increases which leads to inflation and that eventually results in the rise in gold price.

6.6 Jewellery& Other domestic demand

- The price of gold increases during the different festivals and wedding seasons since their demand increases.
- Demand for gold also comes from other domestic manufacturing needs.

7. RELATION OF EACH VARIABLE WITH GOLD PRICE

7.1 USD

The value of gold and dollars act inversely. As the value of the US dollar rises the value of gold falls. Similarly, as the value of the dollar falls, the value of gold jumps higher.

7.2 LMBA

It is the market which sets the gold prices in us dollar, so prices of the gold for world are set by the LMBA.

Gold prices & LBMA are move together as the gold prices are set by the LBMA in us dollar and then gold is imported in India .

7.3 Crude Oil

Crude oil is one of the most important commodities and energy sources, vital for businesses & individual consumers.

Important link between gold and oil is inflation. As crude oil prices rise, inflation also rises. Gold is known to be a good hedge against inflation. The value of gold only increases when inflation rises. Over 60% of the time, gold and crude oil have a direct relationship

7.4 HUI

The **NYSE Arca Gold BUGS Index** better known as the **HUI Gold Index**, is an index of publicly traded gold-mining companies that is useful for tracking short-term trends in gold prices.

This Index currently consists of 15 of the largest and most widely held public gold production companies.

7.5 Sensex

Gold price & sensex have a negative relationship with each other

When the SENSEX goes up, the value of rupee strengthens against the US Dollar and the gold prices fall. The logic behind this is that most of the big-ticket market players invest in shares or gold. When the share market sentiments improve, they shift their investments from the yellow metal to shares.

Gold Price and SENSEX are said to have negative relation but they tends to move in the same direction. That is initially when the gold price was high and was increasing, the SENSEX followed the same path but stayed

below the gold price. Similarly when the SENSEX price rose high, the gold price followed almost similar direction even though with small troughs and peaks, but it was below the SENSEX index.

And during the wake of Covid, when price of gold rose high, SENSEX index went down drastically.

But they tends to follow none of these assumptions lately i.e, from 2021 onwards.

7.6 Nifty

Gold price & nifty have a negative relationship with each other. When the value of the nifty stock prices goes up the prices of the gold goes down.

In the end of year 2013 the rupee lost its value against the us dollar , causing the chaos in the domestic economy , imports became costlier, fuel prices became costlier , there was inflation in the economy and then people started investing In the gold & prices for the gold went up.

Then again we can see a dip in the nifty stocks in the year of 2020 , this was mainly due the pandemic when everything was shut and then gold prices suddenly went up as people starting investing more in the gold .

7.7 S&P

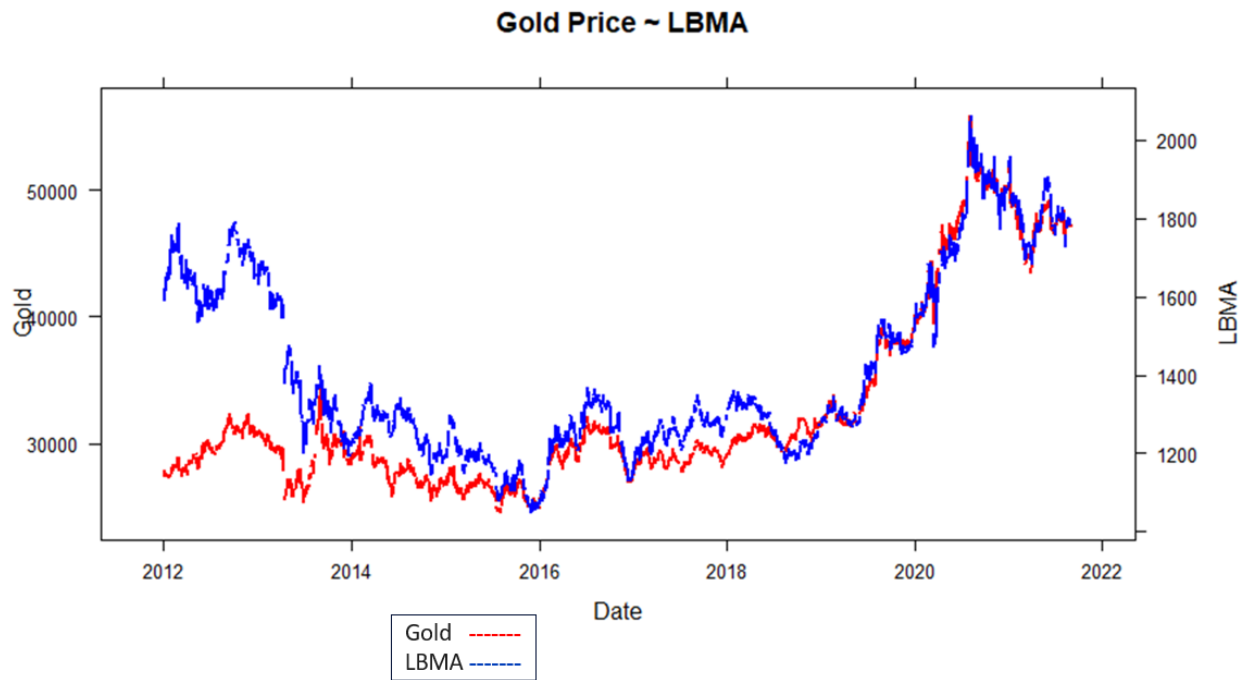
The S&P 500 Index features the 500 largest U.S. publicly traded companies, weighted by market capitalization. The relationship between stock valuations and the gold price is that these 2 markets are negatively linked. when traders go into defensive mode, they may prefer gold to relatively risky stocks.

7.8 INR

Gold and Rupee are inter-dependent. If the Rupee's value increases (in comparison to the dollar), then gold prices go up.

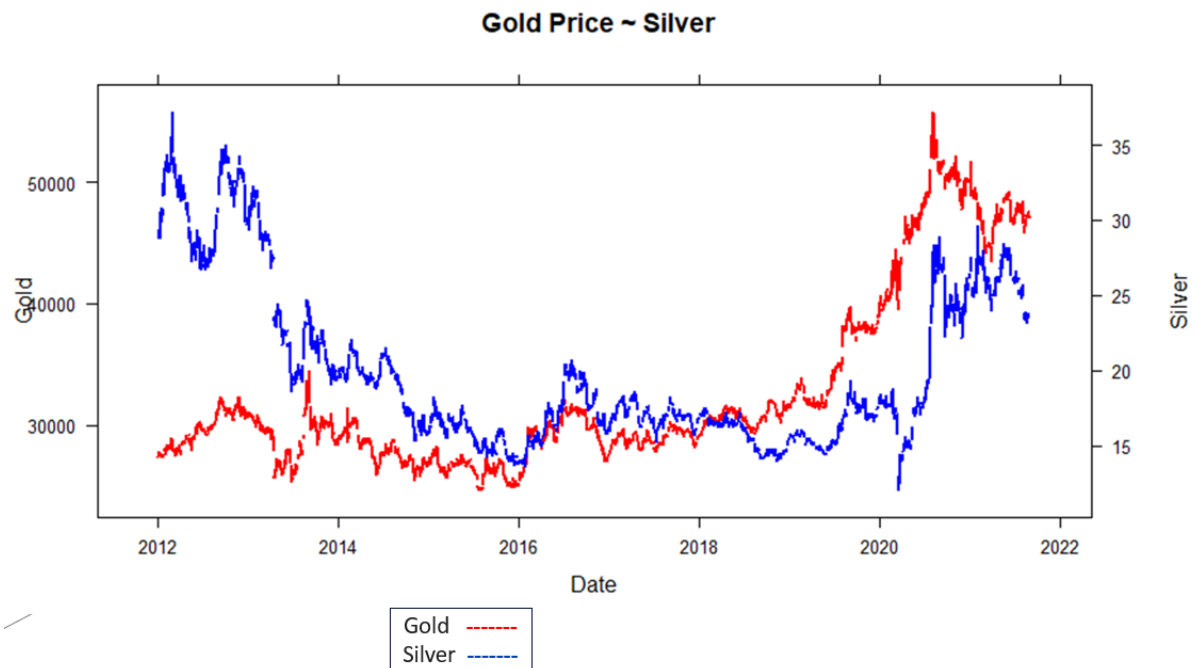
8. EXPLORATORY DATA ANALYSIS

8.1



Gold price & LBMA always move together .

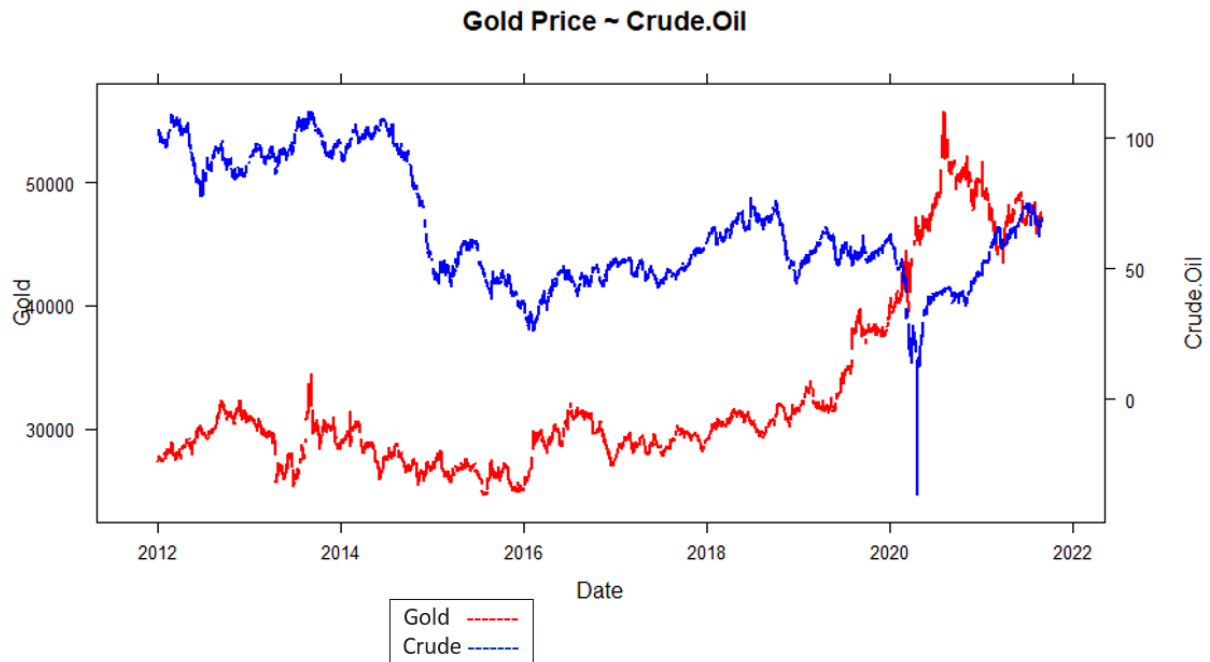
8.2



Both Gold and silver price moves in the same direction mostly, keeping aside the small troughs and peaks or changes that happen in the short term.

During the wake of Covid pandemic, there was a mismatch in their movement. Silver price when down initially and then started rising whereas Gold price rose high.

8.3

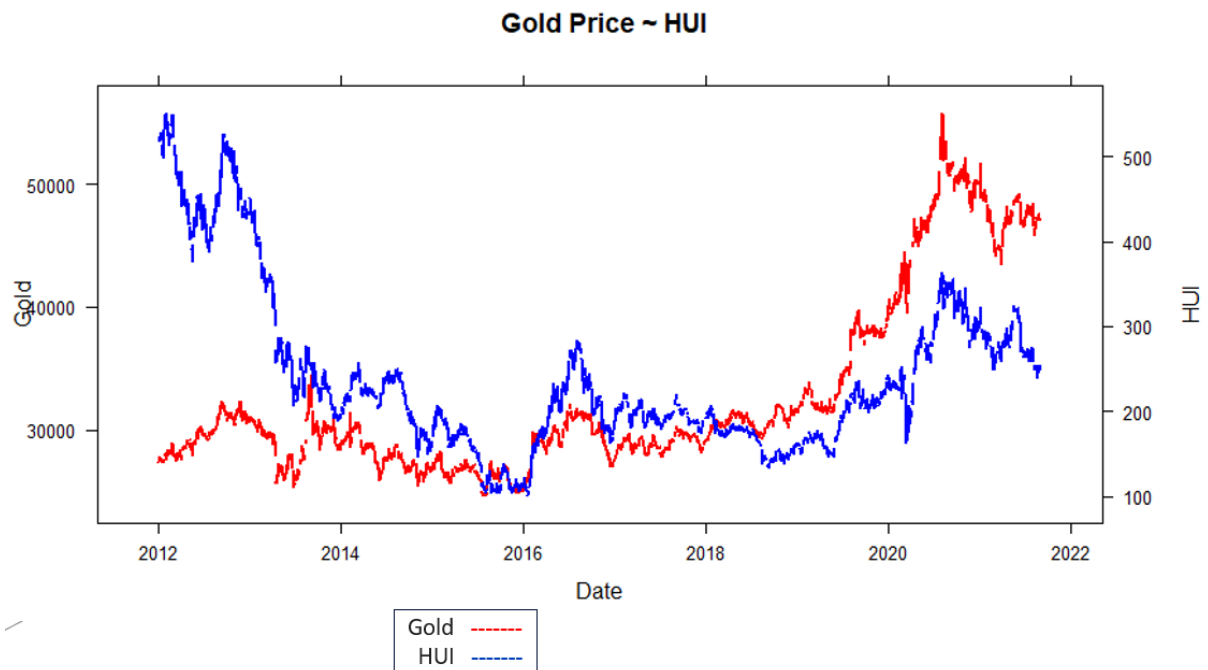


Gold price & crude oil move together, though there is difference in the prices of both commodities.

Both Gold and crude oil move in the same direction mostly, keeping aside the small troughs and peaks or changes that happen in the short term.

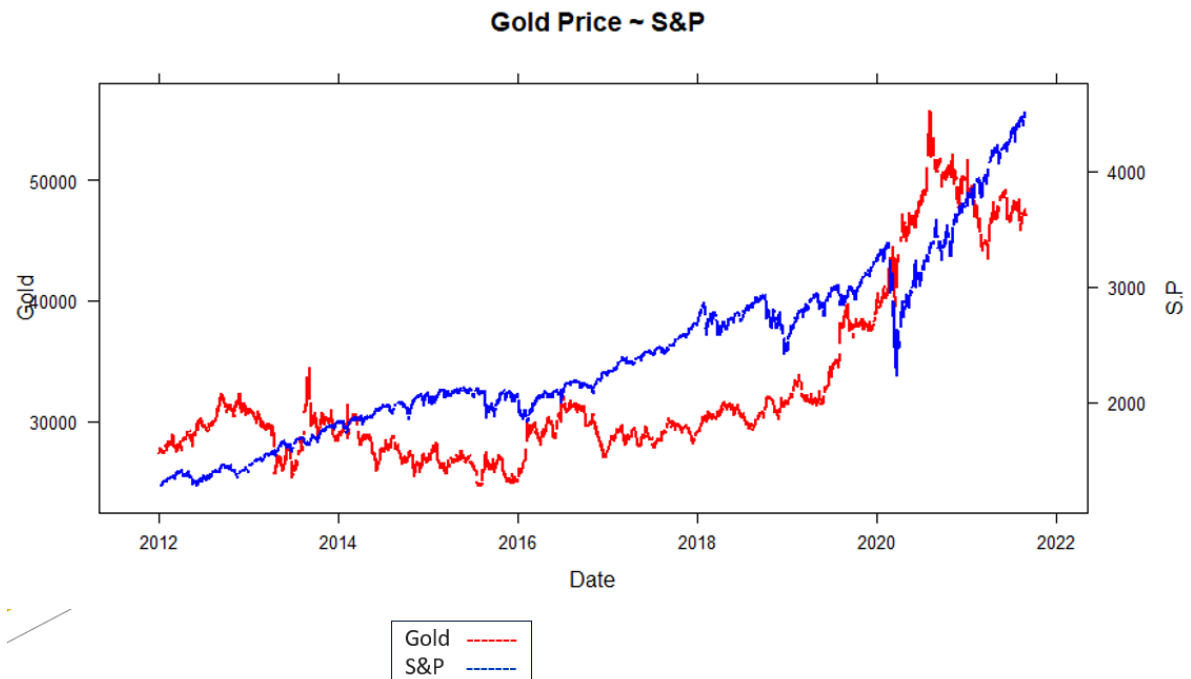
During the wake of Covid pandemic, there was a mismatch in their movement. Crude oil price went down initially and then started rising whereas Gold price rose high.

8.4



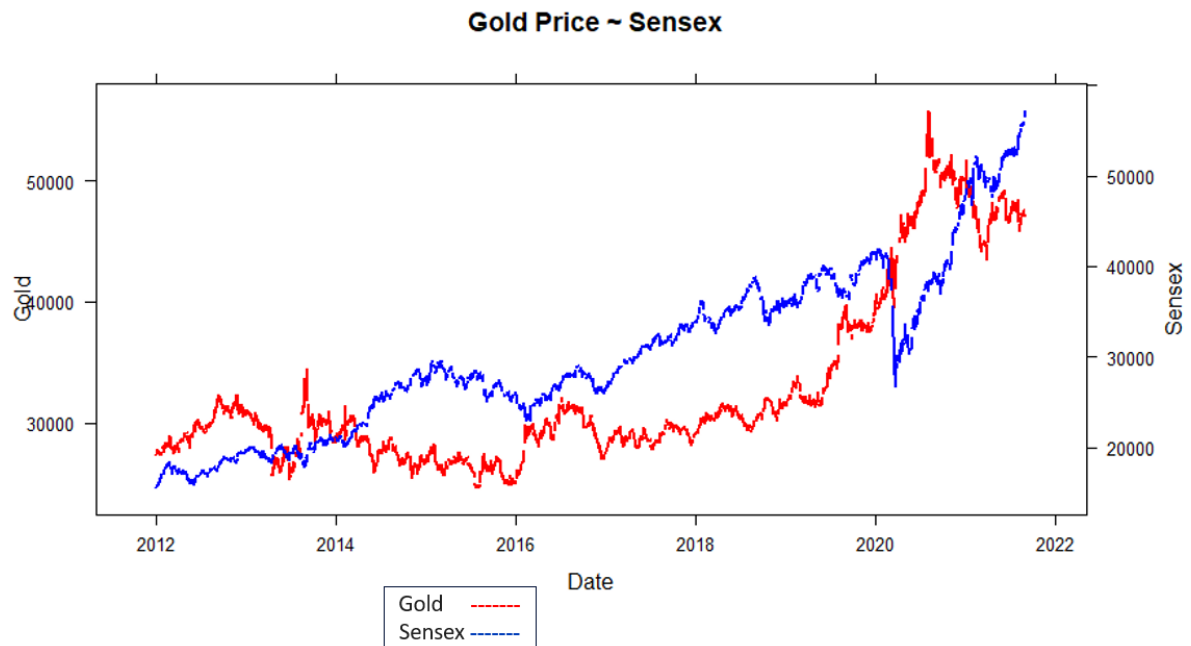
Gold prices & hui index also move together & have a positive correlation. Gold Price and hui are said to have negative relation but they tends to move in the same direction. That is initially when the hui index was high, the gold followed the same path but stayed below the HUI index. Similarly when the Gold price rose high, the HUI index price followed almost similar direction even though with small troughs and peaks, but it was below the gold index.

8.5



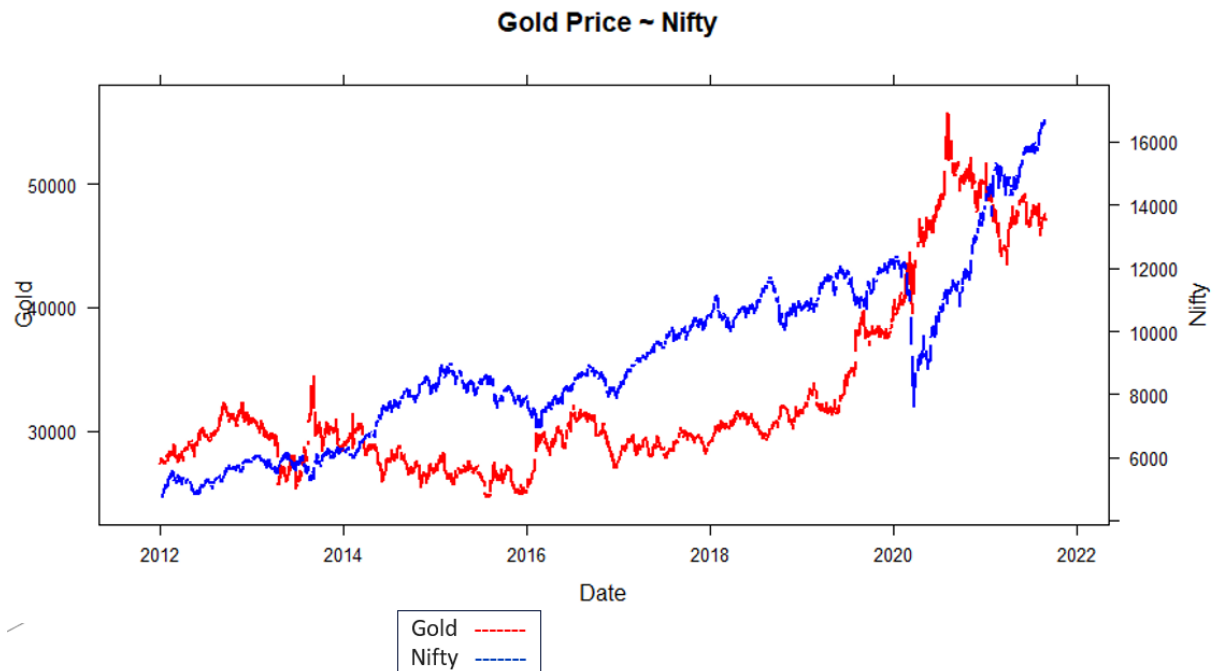
The relationship between stock valuations and the gold price is that these 2 markets are negatively linked. when traders go into defensive mode, they may prefer gold to relatively risky stocks.

8.6



Gold price & sensex have a negative relationship with each other. When the SENSEX goes up, the value of rupee strengthens against the US Dollar and the gold prices fall.

8.7



Gold price & nifty have a negative relationship with each other. When the value of the nifty stock prices goes up the prices of the gold goes down.

8.8



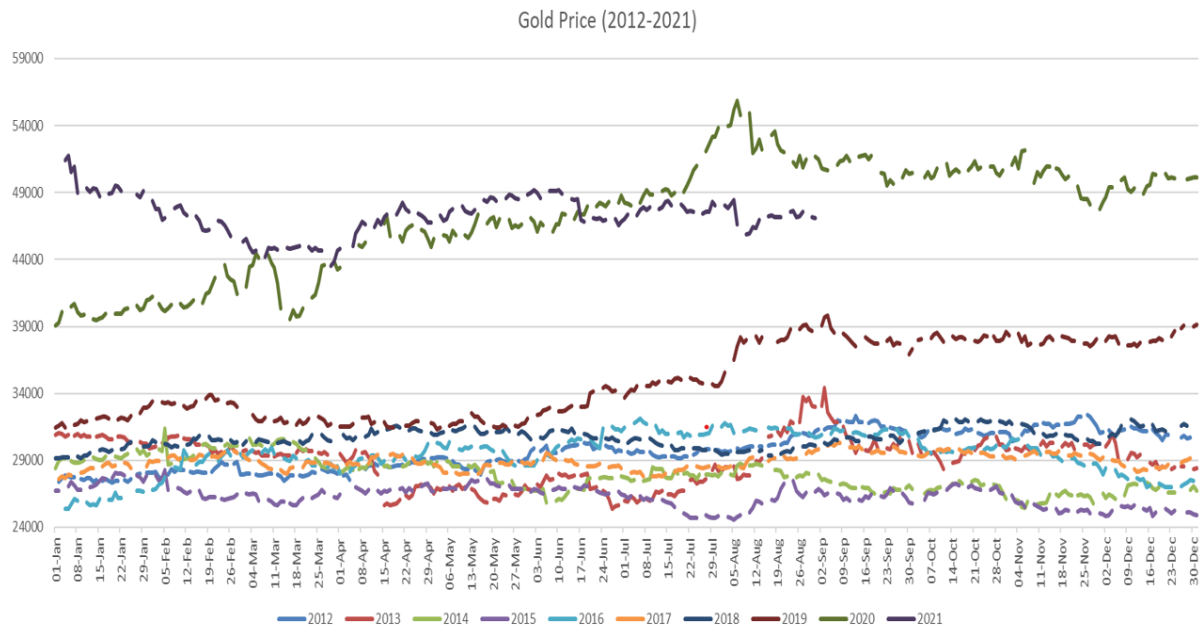
The value of gold and dollars act inversely. As the value of the US dollar rises the value of gold falls. Similarly, as the value of the dollar falls, the value of gold jumps higher.

When the value of us dollar goes down, the value of gold decrease (in dollar)

8.9



Gold and Rupee are inter-dependent. If the Rupee's value increases (in comparison to the dollar), then gold prices go up.



Gold prices have an increasing & decreasing trend, it is because of the festivals (Diwali & Akshaya Tritiya) or the marriages. During these auspicious days people start buying gold & hence prices of gold rise.

9. MACHINE LEARNING ALGORITHMS

1. Simple Exponential Smoothing

The simplest of the exponentially smoothing methods is naturally called **simple exponential smoothing**.

Exponentially smoothing is usually used to make short term forecasts .

2. Holts Winter Exponential Smoothing

Holt's – winter method uses exponential smoothing to encode lots of values from the past & use them to predict values for present & future .

3. Holts Winter Method

Winter's smoothing technique allows us to include seasonality while making the prediction along with the trend.

4. ARIMA

ARIMA stands for autoregressive integrated moving average .Using Arima model we can forecast a time series using the series past values .

5. ARIMAX

ARIMAX or Regression ARIMA is an extension of ARIMA model.

The X added to the end stands for “exogenous”.

In forecasting, this method involves independent variables.

6. Multivariate Regression

Multivariate Regression is a supervised machine learning algorithm involving multiple data variables for analysis. A Multivariate regression is an extension of multiple regression with one dependent variable and multiple independent variables.

7. FB Prophet

Prophet is procedure for forecasting time series data based on an additive model where non-linear trends are fit with yearly ,weekly & daily seasonality, plus holiday effects. It works best with time series that have strong seasonal effects & several seasons of historical data .

10. PREDICTIVE MODELLING

10.1 Simple Exponential Smoothing

```
from pandas import read_csv
import pandas as pd
gold=pd.read_csv("C:/Users/KEERTHIMA/Desktop/data_events.csv")
```

```
gold.head()
```

	Date	Silver	USD	INR	LBMA	Crude	HUI	S&P	Sensex	Nifty	Price	Diwali	Akshaya Trithya
0	01-01-2012	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0	0
1	02-01-2012	28.160	NaN	NaN	NaN	NaN	NaN	NaN	15488.37	NaN	27382.0	0	0
2	03-01-2012	29.533	79.610001	53.11	1590.0	102.96	522.20	1277.06	15797.70	4765.30	27656.0	0	0
3	04-01-2012	29.063	80.089996	52.85	1603.0	103.22	521.44	1277.30	15919.29	4749.65	27852.0	0	0
4	05-01-2012	29.265	80.940002	52.89	1614.5	101.81	522.64	1281.06	15884.91	4749.95	27821.0	0	0

```
type(gold1)
```

```
pandas.core.frame.DataFrame
```

```
len(gold1)
```

```
2586
```

```

from pandas import to_datetime
gold1['Date'] = to_datetime(gold1['Date'])

train=gold1.drop(gold1.index[round(len(gold1)*0.8):])
print(train)
test=gold1.drop(gold1.index[:round(len(gold1)*0.8)])
print(test)

```

	Date	Price
1	2012-02-01	27382.0
2	2012-03-01	27656.0
3	2012-04-01	27852.0
4	2012-05-01	27821.0
5	2012-06-01	27755.0

...	...	...
2790	2019-08-22	38165.0
2791	2019-08-23	38765.0
2794	2019-08-26	38828.0
2795	2019-08-27	39044.0
2796	2019-08-28	39165.0

[2069 rows x 2 columns]

	Date	Price
2797	2019-08-29	38803.0
2798	2019-08-30	38656.0
2801	2019-02-09	39087.0
2802	2019-03-09	39680.0
2803	2019-04-09	39823.0

...	...	...
3524	2021-08-25	47179.0
3525	2021-08-26	47237.0
3526	2021-08-27	47538.0
3529	2021-08-30	47164.0
3530	2021-08-31	47120.0

[517 rows x 2 columns]



```
fittrain= fit2.fittedvalues
truedata= train.Price
import numpy as np
actual = test.Price
pred = y_hat_avg.SES
def mape(actual, pred):
    actual, pred = np.array(actual), np.array(pred)
    return np.mean(np.abs((actual - pred) / actual)) * 100
print(mape(actual, pred))
print(mape(truedata, fittrain))
```

```
14.397996588823636
0.5966570911771938
```

10.2 Holts Winter Exponential Smoothing

```
from pandas import read_csv
import pandas as pd
gold=pd.read_csv("C:/Users/KEERTHIMA/Desktop/data_events.csv")
gold1=gold[["Date","Price"]]
gold1=gold1.dropna()
gold1.head()

from pandas import to_datetime
gold1['Date']= to_datetime(gold1['Date'])

train=gold1.drop(gold1.index[round(len(gold1)*0.8):])
print(train)
test=gold1.drop(gold1.index[:round(len(gold1)*0.8)])
print(test)

from statsmodels.tsa.api import ExponentialSmoothing, SimpleExpSmoothing, Holt
import numpy as np
from matplotlib import pyplot as plt
```

```
      Date      Price
1  2012-02-01  27382.0
2  2012-03-01  27656.0
3  2012-04-01  27852.0
4  2012-05-01  27821.0
5  2012-06-01  27755.0
...      ...      ...
2790 2019-08-22  38165.0
2791 2019-08-23  38765.0
2794 2019-08-26  38828.0
2795 2019-08-27  39044.0
2796 2019-08-28  39165.0
```

```
[2069 rows x 2 columns]
      Date      Price
2797 2019-08-29  38803.0
2798 2019-08-30  38656.0
2801 2019-02-09  39087.0
2802 2019-03-09  39680.0
2803 2019-04-09  39823.0
...      ...      ...
3524 2021-08-25  47179.0
```

```
fit1 = Holt(np.asarray(train['Price'])).fit(smoothing_level = 1,smoothing_slope = 0.002)
y_hat_avg['Holt_linear'] = fit1.forecast(len(test))
fittrain1=fit1.fittedvalues
plt.figure(figsize=(16,8))
plt.plot(train['Price'], label='Train')
plt.plot(fittrain1,label='Fitted train')
plt.plot(test['Price'], label='Test')
plt.plot(y_hat_avg['Holt_linear'], label='Holt_linear')
plt.legend(loc='best')
plt.title('HOLT WINTERS EXPONENTIAL SMOOTHING')
plt.show()
```



```
n [16]: import numpy as np
actual = test.Price
pred = y_hat_avg.Holt_linear
def mape(actual, pred):
    #actual, pred = np.array(actual), np.array(pred)
    return np.mean(np.abs((actual - pred) / actual)) * 100
print(mape(actual, pred))
print(mape(truedata, fittrain1))

6.463987819744865
0.596788338255985
```

10.3 Holts Winter Method

```
from pandas import read_csv
import pandas as pd
gold=pd.read_csv("C:/Users/KEERTHIMA/Desktop/data_events.csv")
gold1=gold[["Date","Price"]]
gold1=gold1.dropna()
gold1.head()

from pandas import to_datetime
gold1['Date']= to_datetime(gold1['Date'])

train=gold1.drop(gold1.index[round(len(gold1)*0.8):])
print(train)
test=gold1.drop(gold1.index[:round(len(gold1)*0.8)])
print(test)

from statsmodels.tsa.api import ExponentialSmoothing, SimpleExpSmoothing, Holt
import numpy as np
from matplotlib import pyplot as plt
```

	Date	Price
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2801	2019-02-09	39087.0
2802	2019-03-09	39680.0
2803	2019-04-09	39823.0
...	...	...
3524	2021-08-25	47179.0
3525	2021-08-26	47237.0


```

fit3 = ExponentialSmoothing(np.asarray(train['Price']), seasonal_periods=12, trend='add', seasonal='add').fit(smoothing_level = 1, smoothing_slope = 0.002)
y_hat_avg['Holt_Winter'] = fit3.forecast(len(test))
fittrain3=fit3.fittedvalues
plt.figure(figsize=(16,8))
plt.plot(train['Price'], label='Train')
plt.plot(fittrain3, label='Fitted train')
plt.plot(test['Price'], label='Test')
plt.plot(y_hat_avg['Holt_Winter'], label='Holt_Winter')
plt.legend(loc='best')
plt.title('HOLT WINTERS METHOD')
plt.show()

```



```

import numpy as np
actual = test.Price
pred = y_hat_avg.Holt_Winter
def mape(actual, pred):
    #actual, pred = np.array(actual), np.array(pred)
    return np.mean(np.abs((actual - pred) / actual)) * 100
print(mape(actual, pred))
print(mape(truedata, fittrain3))

```

```

6.472628847959817
0.5980587303773843

```

```

y_hat_avg

```

	Date	Price	SES	Holt_Winter	Holt_linear
2797	2019-08-29	38803.0	39165.0	39180.126088	39182.273895
2798	2019-08-30	38656.0	39165.0	39212.915231	39199.547789
2801	2019-02-09	39087.0	39165.0	39211.283954	39216.821684
2802	2019-03-09	39680.0	39165.0	39240.854510	39234.095578
2803	2019-04-09	39823.0	39165.0	39217.544181	39251.369473
...	...	...	...	...	...
3524	2021-08-25	47179.0	39165.0	48010.511512	48026.507890
3525	2021-08-26	47237.0	39165.0	48053.531049	48043.781784
3526	2021-08-27	47538.0	39165.0	48057.727712	48061.055679
3529	2021-08-30	47164.0	39165.0	48070.365667	48078.329573
3530	2021-08-31	47120.0	39165.0	48085.491755	48095.603468

517 rows × 5 columns

10.4 ARIMA

```
#1.Data Loading
gold=read.csv("C:/Users/ancha/OneDrive/Desktop/project/data_events1.csv")
gold=gold[,-(2:10)]
g=na.omit(gold)
head(g)
```

	Date <chr>	Price <int>	Diwali <int>	Akshaya.Trithya <int>
2	02-01-2012	27382	0	0
3	03-01-2012	27656	0	0
4	04-01-2012	27852	0	0
5	05-01-2012	27821	0	0
6	06-01-2012	27755	0	0
7	07-01-2012	27762	0	0

6 rows

```
#2.Creating Gold Price data & xreg
```

```
d=g[,1:2]
```

```
x=g[,3:4]
```

```
#3.Splitting Train & Test data:
```

```
train_data=head(d,round(nrow(d)*0.8))
```

```
test_data=tail(d,round(nrow(d)*0.2))
```

```
#4.Fitting the ARIMA model:
```

```
install.packages("forecast")
```

```
library(forecast)
```

```
model=auto.arima(train_data$Price)
```

```
summary(model)
```

```

Series: train_data$Price
ARIMA(2,1,2) with drift

Coefficients:
      ar1      ar2      ma1      ma2      drift
      1.3986  -0.8120  -1.4342   0.8681   5.2545
s.e.    0.0642   0.0707   0.0548   0.0606   6.0240

sigma^2 estimated as 68296:  log likelihood=-14441.98
AIC=28895.96  AICc=28896.01  BIC=28929.77

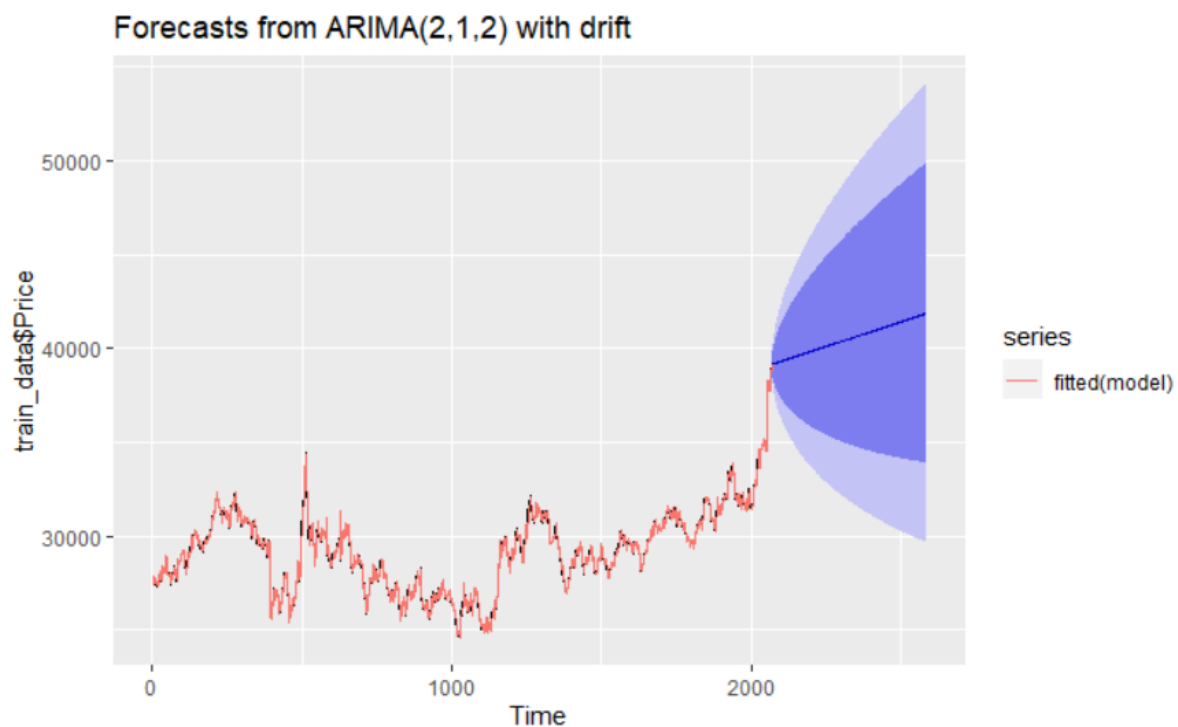
Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.4281852 260.9565 174.0789 -0.004497708 0.5976805 1.000779 -0.02827718

```

```

#5.Forecasting test data from train data
forecast1=forecast(model,h=517)
library(ggplot2)
autoplot(forecast1)
autoplot(forecast1)+autolayer(fitted(model))

```



#6.Validating the model

```
accuracy(forecast1,test_data$Price)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.4281852	260.9565	174.0789	-0.004497708	0.5976805	1.000779	-0.02827718
Test set	5229.6841712	6665.6574	5675.7211	10.595449156	11.7702332	32.629696	NA

#6.Computing MAPE of Test data

```
library(MLmetrics)
pred=forecast1$mean
m=MAPE(pred,test_data$Price)
m*100 #=11.77%
```

```
[1] 11.77023
```

10.5 Arimax

```
#4.Creating Train & Test of xreg
y=as.data.frame(ifelse((x$Diwali) == 1 | (x$Akshaya.Trithya)==1,1,0))
names(y)="y"
View(y)
ytrain=head(y,round(nrow(y)*0.8))
ytest=tail(y,round(nrow(y)*0.2))
```

```
#5.Converting xreg data to matrix
ytrainm=as.matrix(ytrain)
ytestm=as.matrix(ytest)
```

```
#6.Fitting ARIMAX model
modelx=auto.arima(train_data$Price,xreg = ytrainm)
summary(modelx)
```

```
Series: train_data$Price
Regression with ARIMA(2,1,3) errors
```

```
Coefficients:
```

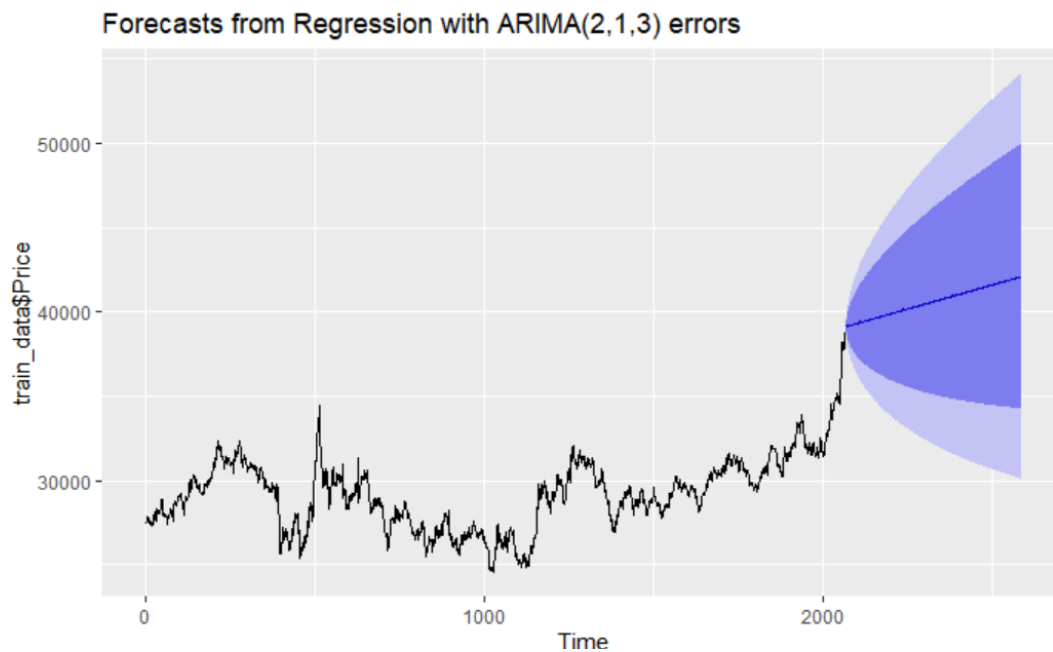
	ar1	ar2	ma1	ma2	ma3	drift	y
	1.4277	-0.8040	-1.4876	0.9138	-0.0386	5.7107	63.6355
s.e.	0.0818	0.0795	0.0857	0.0884	0.0277	5.9051	48.9517

```
sigma^2 estimated as 68232: log likelihood=-14440
AIC=28896 AICc=28896.07 BIC=28941.08
```

```
Training set error measures:
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	-0.01229985	260.7074	174.0692	-0.006119519	0.5976515	1.000723	-0.003466662

```
#7.Forecasting
forecastx=forecast(modelx,xreg = ytestm)
autoplot(forecastx)
```



```
#8.Validating model
accuracy(forecastx,test_data$Price)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	-0.01229985	260.7074	174.0692	-0.006119519	0.5976515	1.000723	-0.003466662
Test set	5123.35157825	6558.3455	5573.8116	10.371992922	11.5582609	32.043819	NA

```
#Computing MAPE
predx=forecastx$mean
library(MLmetrics)
mx=MAPE(predx,test_data$Price)
mx*100
```

```
[1] 11.55826
```

10.6 Multivariate Regression

```
##{r}
d1<-read.csv('C:/Users/ancha/OneDrive/Desktop/project/Data1.csv')
d1
d2<-na.omit(d1)
View(d2)
d3<-d2[,-1]
d3
##
```

Description: df [2,220 x 11]

	Date <chr>	Silver <dbl>	USD <dbl>	INR <dbl>	LBMA <dbl>	Crude <dbl>	HUI <dbl>	S.P <dbl>	Sensex <dbl>
3	03-01-2012	29.533	79.61	53.11	1590.00	102.96	522.20	1277.06	15797.70
4	04-01-2012	29.063	80.09	52.85	1603.00	103.22	521.44	1277.30	15919.29
5	05-01-2012	29.265	80.94	52.89	1614.50	101.81	522.64	1281.06	15884.91
6	06-01-2012	28.653	81.24	52.67	1621.00	101.56	517.66	1277.81	15830.76
9	09-01-2012	28.749	81.05	52.39	1618.00	101.31	519.76	1280.70	15801.19
10	10-01-2012	29.783	80.80	51.59	1627.00	102.24	527.76	1292.08	16035.68
11	11-01-2012	29.858	81.35	51.78	1641.00	100.89	522.03	1292.48	16192.68
12	12-01-2012	30.092	80.77	51.51	1652.50	99.03	526.74	1295.50	16073.97
13	13-01-2012	29.493	81.43	51.43	1642.00	98.69	519.39	1289.09	16151.58
17	17-01-2012	30.106	81.18	50.69	1662.00	100.70	511.51	1293.67	16377.29

1-10 of 2,220 rows | 1-10 of 11 columns

Previous 1 2 3 4 5 6 ... 100 Next

```
##{r}
m1<-lm(Price ~Silver+USD+INR+LBMA+Crude+HUI+S.P+Sensex+Nifty, data =d3)
summary(m1)
##
```

Call:

```
lm(formula = Price ~ Silver + USD + INR + LBMA + Crude + HUI +
    S.P + Sensex + Nifty, data = d3)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-2839.6  -429.1   -26.6    432.2   3360.5
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.717e+04  8.423e+02 -20.385  <2e-16 ***
Silver       -9.498e+00  9.978e+00  -0.952    0.341
USD          -1.455e+02  6.165e+00 -23.593  <2e-16 ***
INR           5.131e+02  7.983e+00  64.280  <2e-16 ***
LBMA          2.197e+01  3.316e-01  66.261  <2e-16 ***
Crude        -2.869e+01  1.760e+00 -16.300  <2e-16 ***
HUI           -7.148e+00  7.052e-01 -10.136  <2e-16 ***
S.P           4.024e+00  1.669e-01  24.110  <2e-16 ***
Sensex        5.184e-01  4.244e-02  12.215  <2e-16 ***
Nifty        -2.520e+00  1.495e-01 -16.855  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
install.packages("tidyverse")
library(tidyverse)
install.packages("caret")
library(caret)
install.packages("forecast")
library(forecast)
```

```
##{r}
summary(m1)$coef
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.717018e+04	842.2895441	-20.3851327	8.776627e-85
Silver	-9.498147e+00	9.9777505	-0.9519327	3.412352e-01
USD	-1.454586e+02	6.1652036	-23.5934859	5.897995e-110
INR	5.131390e+02	7.9829087	64.2796964	0.000000e+00
LBMA	2.197086e+01	0.3315801	66.2610927	0.000000e+00
Crude	-2.868798e+01	1.7599514	-16.3004375	1.694552e-56
HUI	-7.148471e+00	0.7052302	-10.1363645	1.241268e-23
S.P	4.024471e+00	0.1669184	24.1104067	3.198161e-114
Sensex	5.184084e-01	0.0424415	12.2146578	2.986566e-33
Nifty	-2.519611e+00	0.1494864	-16.8551167	4.618702e-60

```
##{r}
str(m1)
```

```
List of 12
 $ coefficients : Named num [1:10] -17170.2 -9.5 -145.5 513.1 22 ...
 ..- attr(*, "names")= chr [1:10] "(Intercept)" "Silver" "USD" "INR" ...
 $ residuals    : Named num [1:2220] -136 -128 -335 -384 -395 ...
 ..- attr(*, "names")= chr [1:2220] "3" "4" "5" "6" ...
 $ effects      : Named num [1:2220] -1541195 -92995 170866 264171 -102676 ...
 ..- attr(*, "names")= chr [1:2220] "(Intercept)" "Silver" "USD" "INR" ...
 $ rank         : int 10
 $ fitted.values: Named num [1:2220] 27792 27980 28156 28139 27973 ...
 ..- attr(*, "names")= chr [1:2220] "3" "4" "5" "6" ...
 $ assign       : int [1:10] 0 1 2 3 4 5 6 7 8 9
 $ qr           :List of 5
 ..$ qr        : num [1:2220, 1:10] -47.1169 0.0212 0.0212 0.0212 0.0212 ...
 .. ..- attr(*, "dimnames")=List of 2
 .. .. ..$ : chr [1:2220] "3" "4" "5" "6" ...
 .. .. ..$ : chr [1:10] "(Intercept)" "Silver" "USD" "INR" ...
 .. ..- attr(*, "assign")= int [1:10] 0 1 2 3 4 5 6 7 8 9
 ..$ qraux: num [1:10] 1.02 1.03 1.01 1.03 1.02 ...
```

```
##{r}
saml<-sample.int(dim(d3)[1],0.7*dim(d3)[1])
datatrain=d3[saml,]
datatest=d3[-saml,]
View(datatrain)
View(datatest)
str(datatest)
```

```
'data.frame': 666 obs. of 11 variables:
 $ Date : chr "05-01-2012" "10-01-2012" "12-01-2012" "13-01-2012" ...
 $ Silver: num 29.3 29.8 30.1 29.5 32.2 ...
 $ USD : num 80.9 80.8 80.8 81.4 79.7 ...
 $ INR : num 52.9 51.6 51.5 51.4 50 ...
 $ LBMA : num 1614 1627 1652 1642 1675 ...
 $ Crude : num 101.8 102.2 99 98.7 99.5 ...
 $ HUI : num 523 528 527 519 509 ...
 $ S.P : num 1281 1292 1296 1289 1316 ...
 $ Sensex: num 15885 16036 16074 16152 16716 ...
 $ Nifty : num 4750 4850 4831 4866 5046 ...
 $ Price : int 27821 27537 27665 27427 27510 27711 28373 28274 27952 27847 ...
```



```
{r}
names(d3)<-c("Date","Silver","USD","INR","LBMA","Crude","HUI","S.P","Sensex","Nifty","Price" )
View(d3)
m1<-lm(Price ~Silver+USD+INR+LBMA+Crude+HUI+S.P+Sensex+Nifty,data =datatrain)
summary(m1)
```

```
Call:
lm(formula = Price ~ Silver + USD + INR + LBMA + Crude + HUI +
    S.P + Sensex + Nifty, data = datatrain)
```

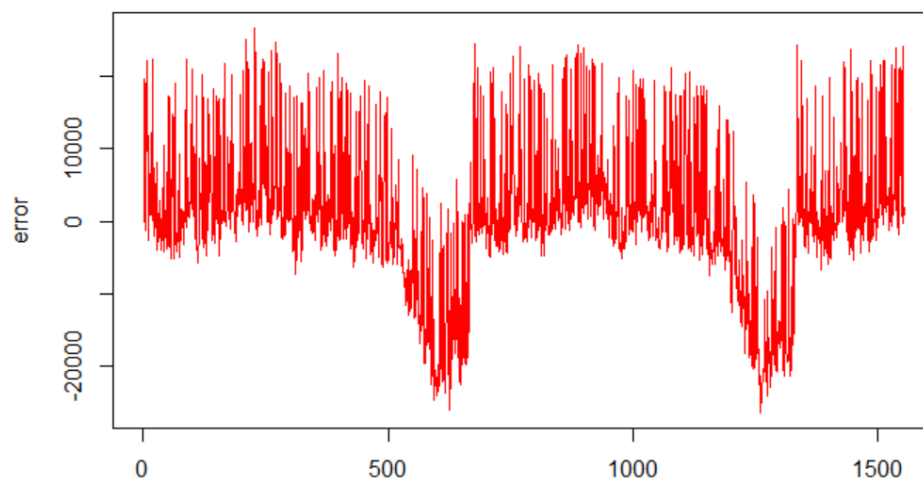
```
Residuals:
    Min       1Q   Median       3Q      Max
-2809.76  -430.79   -30.93   438.78  2803.03
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.774e+04  9.862e+02 -17.989  <2e-16 ***
Silver      -1.455e+01  1.178e+01  -1.235    0.217
USD         -1.395e+02  7.297e+00 -19.113  <2e-16 ***
INR          5.108e+02  9.524e+00  53.633  <2e-16 ***
LBMA         2.202e+01  3.909e-01  56.331  <2e-16 ***
Crude       -2.726e+01  2.076e+00 -13.129  <2e-16 ***
HUI         -6.856e+00  8.254e-01  -8.306  <2e-16 ***
S.P          3.970e+00  1.971e-01  20.139  <2e-16 ***
Sensex       5.206e-01  5.004e-02  10.403  <2e-16 ***
Nifty       -2.508e+00  1.771e-01 -14.160  <2e-16 ***
```

```
{r}
m2=forecast(datatrain,h=666)
predictions<-m1 %>% predict(datatrain)
View(predictions)
predictions1<-forecast(m1,newdata = datatest)
accuracy(predictions1,datatest)
```

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	-0.00000000000003497953	649.7845	514.5514	-0.01875251	1.615733	0.09000399
Test set	24.15659565625735893946	669.7269	523.4261	0.05541198	1.640390	0.09155633

```
error<-predictions-datatest$Price
plot(error,type='l',col='red')
```



```

##{r}
install.packages("MLmetrics")
library(MLmetrics)
MAPE(predictions,datatest$Price)
##

```

[1] 0.2086688

```

#with lagged values
dynlm
install.packages("dynlm")
library(dynlm)
md<-dynlm(Price~L(LBMA, 1) + L(HUI, 1)+L(S.P, 1)+L(Sensex, 1)+
          L(Nifty, 1),data=datatrain)

options(scipen = 999)
summary(md)

```

Time series regression with "zoo" data:
Start = 1, End = 1554

Call:
dynlm(formula = Price ~ L(LBMA, 1) + L(HUI, 1) + L(S.P, 1) +
L(Sensex, 1) + L(Nifty, 1), data = datatrain)

Residuals:

	Min	1Q	Median	3Q	Max
	-4846.2	-818.2	24.7	703.1	4580.0

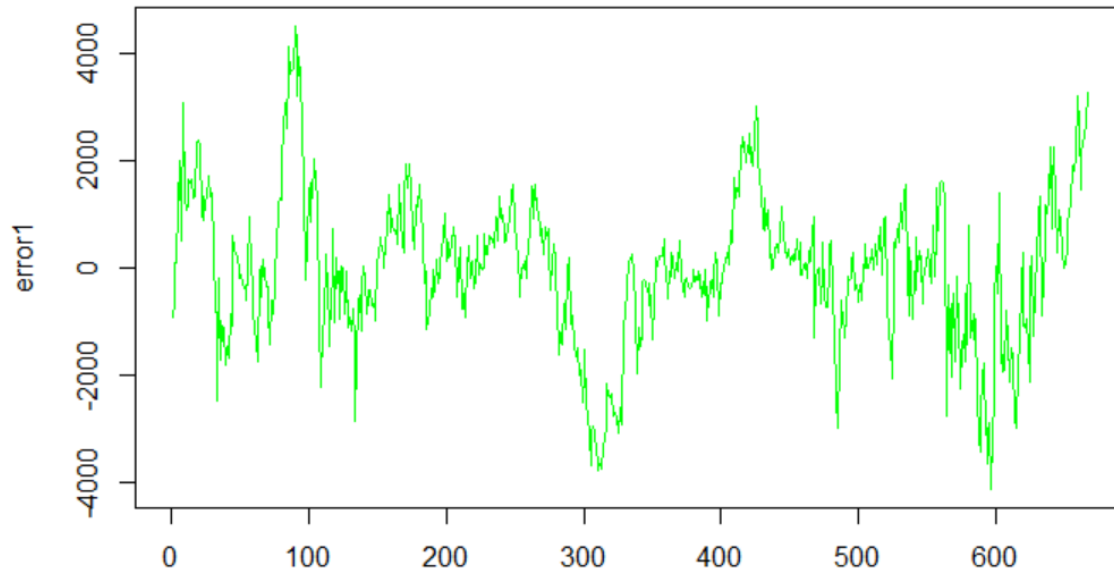
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5759.30417	403.32537	-14.280	<0.0000000000000002 ***
L(LBMA, 1)	27.35577	0.63433	43.126	<0.0000000000000002 ***
L(HUI, 1)	-29.65176	1.32217	-22.427	<0.0000000000000002 ***
L(S.P, 1)	9.30641	0.31439	29.601	<0.0000000000000002 ***
L(Sensex, 1)	0.81701	0.08323	9.816	<0.0000000000000002 ***
L(Nifty, 1)	-4.40086	0.27830	-15.814	<0.0000000000000002 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1331 on 1548 degrees of freedom
Multiple R-squared: 0.9676, Adjusted R-squared: 0.9674
F-statistic: 9232 on 5 and 1548 DF, p-value: < 0.00000000000000022

```
error1<-(predictions1-datatest$Price)
plot(error1,col='green',type='l')
```



```
{r}  
MAPE(predictions1,datatest$Price)  
...
```

```
[1] 0.03094126
```

10.7 FB Prophet

```
#-----FB-PROPHET-----
import fbprophet
import numpy as np
import pandas as pd
from pandas import read_csv
from pandas import to_datetime
from pandas import DataFrame
from fbprophet import Prophet
from sklearn.metrics import mean_absolute_error
from matplotlib import pyplot

path = 'C:/Users/ancha/OneDrive/Desktop/project/data_events1.csv'
df = read_csv(path, header=0)
df1 = df[["Date", "Price"]]
type(df1)
df1.info()
```

```
In [6]: df1.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3531 entries, 0 to 3530
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Date    3531 non-null      object
1   Price   2586 non-null      float64
dtypes: float64(1), object(1)
memory usage: 55.3+ KB
```

In [7]:

```
#Prepare expected columns:
df1.columns = ['ds', 'y']
df1['ds'] = to_datetime(df1['ds'])
```

In [9]: df1.columns = ['ds', 'y']

```
In [10]: df1['ds'] = to_datetime(df1['ds'])
<ipython-input-10-a1741b9a926f>:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df1['ds'] = to_datetime(df1['ds'])
```

```

...: holidays=pd.concat((Diwali,AkshayaTrithya))
...: print(holidays)

```

	holiday	ds	lower_window	upper_window
0	Diwali	2012-11-13	-7	7
1	Diwali	2013-11-03	-7	7
2	Diwali	2014-10-22	-7	7
3	Diwali	2015-11-11	-7	7
4	Diwali	2016-10-30	-7	7
5	Diwali	2017-10-19	-7	7
6	Diwali	2018-11-08	-7	7
7	Diwali	2019-10-27	-7	7
8	Diwali	2020-11-14	-7	7
9	Diwali	2021-11-04	-7	7
0	AkshayaTrithya	2012-04-24	0	0
1	AkshayaTrithya	2013-05-13	0	0
2	AkshayaTrithya	2014-05-02	0	0
3	AkshayaTrithya	2015-04-21	0	0
4	AkshayaTrithya	2016-05-09	0	0
5	AkshayaTrithya	2017-04-28	0	0
6	AkshayaTrithya	2018-04-18	0	0
7	AkshayaTrithya	2019-05-07	0	0
8	AkshayaTrithya	2020-04-26	0	0
9	AkshayaTrithya	2021-05-14	0	0

#Fitting the model:

```

m=Prophet(holidays=holidays)
m.add_country_holidays(country_name='India')
m.fit(train)
m.train_holiday_names#to view the Indian holidays considered

```

```

In [16]:
...: m=Prophet(holidays=holidays)
...: m.add_country_holidays(country_name='India')
...: m.fit(train)
...: m.train_holiday_names #to view the Indian holidays considered
INFO:numexpr.utils:NumExpr defaulting to 8 threads.
INFO:fbprophet:Disabling daily seasonality. Run prophet with daily_seasonality=True to
override this.
C:\Users\ancha\anaconda3\lib\site-packages\fbprophet\hdays.py:259: Warning: We only support
Diwali and Holi holidays from 2010 to 2030
  warnings.warn(warning_msg, Warning)
Out[16]:
0                Diwali
1            AkshayaTrithya
2                Republic Day
3            Independence Day
4            Gandhi Jayanti
5            Maundy Thursday
6                Holi
7                Mawlid
8            Eid al-Fitr
9            Feast of the Sacrifice
10           New Year's Day
11                Palm Sunday
12                Good Friday
13                Easter Sunday
14            Feast of Pentecost
15    Fest of St. Theresa of Calcutta
16            Feast of the Blessed Virgin
17                All Saints Day
18                All Souls Day
19                Christmas Day
20                Boxing Day
21            Feast of Holy Family
22                Day of Ashura
dtype: object

```

```

#Forecasting:
future1=df1["ds"]
future1=DataFrame(future1)
future1.columns=['ds']
future1['ds']=to_datetime(future1['ds'])
future1
forecast1=m.predict(future1)

```

```
In [21]: future1
Out[21]:
      ds
0  2012-01-01
1  2012-02-01
2  2012-03-01
3  2012-04-01
4  2012-05-01
...
3526 2021-08-27
3527 2021-08-28
3528 2021-08-29
3529 2021-08-30
3530 2021-08-31

[3531 rows x 1 columns]

In [22]: forecast1=m.predict(future1)
C:\Users\ancha\anaconda3\lib\site-packages\fbprophet\hdays.py:259: Warning: We only support
Diwali and Holli holidays from 2010 to 2030
  warnings.warn(warning_msg, Warning)
C:\Users\ancha\anaconda3\lib\site-packages\fbprophet\hdays.py:259: Warning: We only support
Diwali and Holli holidays from 2010 to 2030
  warnings.warn(warning_msg, Warning)

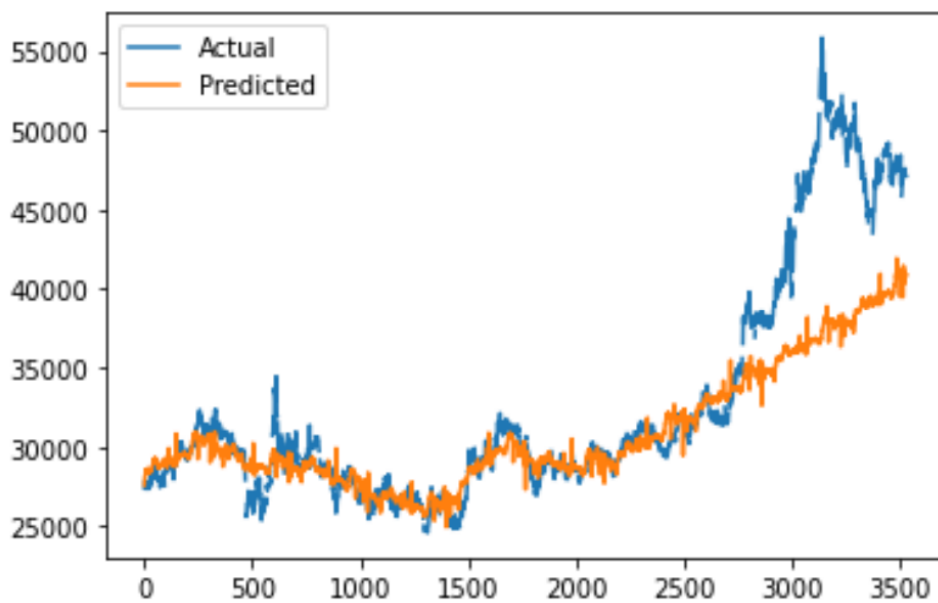
In [23]:
```

```
#Validating
true=df1["y"].values
pred=forecast1['yhat'].values
y0=DataFrame(data=[true,pred]).T
y0.columns=['true','pred']
y0.head()
mape=np.mean(np.abs((y0['true']-y0['pred'])/y0['true']))*.100
print('MAPE: %.3f' % mape) #MAPE: 5.670 (overall MAPE)
```

```
#Train data MAPE
pred=y0.drop(y0.index[round(len(y0)*0.8):])
pred.head()
m1=np.mean(np.abs((train['y']-pred['pred'])/pred['pred']))*100
print('MAPE: %.3f' % m1) #MAPE: 2.755
```

```
In [25]:
...: pred=y0.drop(y0.index[round(len(y0)*0.8):])
...: pred.head()
...: m1=np.mean(np.abs((train['y']-pred['pred'])/pred['pred']))*100
...: print('MAPE: %.3f' % m1) #MAPE: 2.755
MAPE: 2.755
```

```
#Plot forecasted values:  
pyplot.plot(y0['true'], label='Actual')  
pyplot.plot(y0['pred'], label='Predicted')  
pyplot.legend()  
pyplot.show()
```



11.RESULTS AND DISCUSSION

To assess the accuracy of our regression models' predictions, we have relied on the accuracy metric Mean Absolute Error Percentage (MAPE).

The following table summarizes the accuracy result of our models.

Model	Training MAPE	Testing MAPE
Simple Exponential Smoothing	0.596657	14.39799
Holt's Winter Exponential Smoothing	0.596788	6.463987
Holt's Winters	0.598058	6.47262
Multivariate Regression	1.613585	1.644164
Arima	0.5976805	11.7702332
Arimax	0.5976515	11.5582609
FB Prophet	2.755	5.670

12. CONCLUSION

Gold has always occupied a predominant place in the world's economies and among populations. Its importance has only increased since it is still used as a safe haven and as a hedging tool in times of uncertainty and risk.

In this project, we studied the relationship between the gold prices and various other variables that play key roles in the Indian economy. Furthermore, we have attempted to build predictive models based on regression and time series analysis. Thus, we used Simple Exponential Smoothing method, Holt's Exponential method, Holt's Winters method, ARIMA, ARIMAX and Facebook's Prophet method. We also tried Random forest regression model, which we later realized is not suitable for our prediction. Besides, we also attempted LSTM Neural network model but we couldn't make a conclusion out of it since its methods and results were far beyond our knowledge and of advanced level. From our results, we found that most of our models are overfitting even after being treated for overfitting. We assume the drastic changes in gold prices due to the COVID pandemic is the reason behind overfitting of our models. Hence, based on our results we would like to conclude that the FB Prophet model performed better compared to the other models' performance. However, due to the limitations of the model and the complexities in the real world which cannot be fully accurately reflected and forecasted, it is advisable that consumers and investors do a thorough research of the market price movements for making decisions regarding buying or investing in gold.

13. BIBLIOGRAPHY

<https://economictimes.indiatimes.com/wealth/invest/5-factors-that-influence-price-of-gold/articleshow/82341650.cms>
<https://www.bankbazaar.com/gold-rate/top-5-factors-that-affect-gold-rate-in-india.html>
<https://finance.yahoo.com/quote/%5EHUI/history?p=%5EHUI>
<https://fred.stlouisfed.org/graph/?g=4sS1>
<https://www.macrotrends.net/1329/us-dollar-index-historical-chart>
<https://www.federalreserve.gov/datadownload/Download.aspx?rel=H10&series=3604f6e0c01fd1a3275b2de1f6547cb6&filetype=csv&label=include&layout=seriescolumn&from=01/01/1980&to=09/07/2021>
https://www.moneycontrol.com/stocks/hist_index_result.php?indian_indices=4&pno=3&hdn=daily&fdt=2000-09-04&todt=2021-09-04
https://in.investing.com/commodities/refined-gold-historical-data?cid=49776&end_date=1631471400&st_date=853180200
<https://www.lbma.org.uk/prices-and-data/precious-metal-prices#/table>
<https://finance.yahoo.com/quote/%5EYHOH715/history?period1=315532800&period2=1630454400&interval=1d&filter=history&frequency=1d&includeAdjustedClose=true>
https://in.investing.com/commodities/silver-historical-data?end_date=789762600&st_date=600460200
<https://www.thebalance.com/the-relationship-between-gold-and-the-u-s-dollar-808978>
<https://www.analyticsvidhya.com/blog/2018/02/time-series-forecasting-methods/>
<https://www.alpharithms.com/calculating-moving-averages-in-python-585117/>
<https://machinelearningmastery.com/train-test-split-for-evaluating-machine-learning-algorithms/>
<https://www.statology.org/mape-python/>

